**Question 1:** A random variable $X$ has the pdf

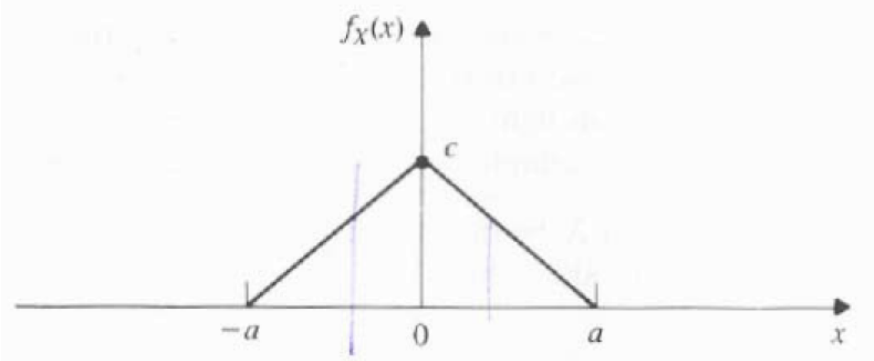
$$f_X(x) = \begin{cases} c(1-x^4), & \text{if } -1 \le x \le 1 \\ 0, & \text{otherwise.} \end{cases}$$

A. Find$c$.

B. Find the cdf of $X$.

C. Find $P[|X| < 1/2]$.

**Question 2:** A random variable $X$ has the pdf shown in the following figure:




A. Find$f_X(x)$.

B. Find the cdf of $X$.

C. Find $b$ such that $P[|X| < b] = 1/2$.

**Question 3:** The cdf of the random variable $X$ is given by

$$f_X(x) = \begin{cases} 1/3 + (2/3)(x+1)^2, & \text{if } -1 \le x \le 0 \\ 0, & \text{if } x < -1. \end{cases}$$

Find the probability of the events $A = X > 1/3, B = |X| \ge 1, C = |X - 1/3| < 1, D = X < 0$.

**Question 4:** The Rayleigh random variable $R$ has the cdf

$$F_R(r) = \begin{cases} 0, & \text{if } r < 0 \\ 1 - e^{-r^2/2\sigma^2}, & \text{if } r \ge 0. \end{cases}$$

Find $P[\sigma \le R \le 2\sigma]$ and $P[R > 3\sigma]$.

**Question 5:** An urn contains 10 chips numbered from 0 to 9. Two chips are drawn at random and without replacement. What is the pmf of their total?

**Question 6:** From the set of families with three children, a family is selected at random and the number of its boys is denoted by the random variable $X$. Find the pmf and the pdf of $X$. Assume that in a three child family, all gender distributions are equally likely.